

SfS-GA: Reconstructing 3D Axially Symmetric Pottery Using a Genetic Algorithm

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Abstract—Computational reassembly of broken archaeological pottery is a combinatorial optimization problem involving a vast search space and a high prevalence of false-positive matches. We present SfS-GA, a 3D pottery reassembly system that uses a genetic algorithm to reconstruct individual axially symmetric pots from their sherds. Building on the Structure-from-Sherds++ (SfS++) pipeline, we replace its incremental beam search with a population-based evolutionary search. SfS-GA uses an integer chromosome representation over a precomputed set of pairwise matched candidates (pairs of sherds). It includes specialised crossover and mutation operators, along with a fitness function incorporating geometric coherence, connectivity, and voxel-based collision detection. Our system was evaluated on the SfS++ benchmark dataset, consisting of 3D scans of real-world archaeological pottery fragments. SfS-GA achieved perfect 100% reconstruction accuracy on manageable combinatorial spaces (Pots A, B, C, F, and G) in a matter of minutes. However, for massively fractured pots and pots that exhibit significant reconstruction ambiguity, i.e., cases where the sherds admit multiple plausible reassemblies, the runtime increases non-linearly, with the evolutionary search challenged by local geometric optima and bounded by the upstream pairwise matching. This paper presents key insights into why genetic algorithms struggle on such pots, exposing the bottlenecks of current feature extraction pipelines. Ultimately, our findings show the strong potential of evolutionary algorithms as an effective, globally aware optimisation framework for the automated restoration of cultural heritage artifacts.

Index Terms—genetic algorithm, pottery reassembly, evolutionary computation, 3D reconstruction, archaeological fragment assembly, fitness function design, combinatorial optimisation

I. INTRODUCTION

Reassembling broken pottery found from archaeological sites is a long standing challenge. Studying these artefacts is important to understand societies that once inhabited these sites. However, even skilled archaeologists and restoration experts, with knowledge of the archaeological context and geometrical structure, find it a tedious and time-consuming process, taking days or even weeks to complete a single pot [1].

This raises a need for computer-driven methods that can solve the problem more efficiently. 3D pottery reconstruction, by nature, is a combinatorial problem. Even a single pot, broken into 10 pieces, has 3,628,800 possible arrangements, not accounting for the rotational and translational degrees of freedom of every sherd. Over the past few years, researchers have devoted considerable effort to this task, with a branch focused on axially symmetric pottery [2].

Present methods range from geometric feature matching to deep learning and parallel search methods, each with its limitations concerning scalability, data requirements, and computational cost [1]. Among the most successful recent methods is Structure-from-Sherds++ (SfS++) [2], which extracts axis-based geometric features and breakline descriptors from each sherd, performs a pairwise matching through a Longest Common Subsequence (LCS) algorithm, and then assembles the pot incrementally using a multi-graph beam search technique called as BAISER (Base-Agnostic Incremental ShErd Registration). This system produces state-of-the-art results on a real pottery dataset, including the simultaneous reconstruction of multiple pots from a collection of mixed fragments.

Despite the breadth of these approaches, the application of evolutionary algorithms, which have shown tremendous success as massive 2D puzzle solvers, remains highly underexplored for 3D pottery. While some isolated efforts have introduced genetic algorithms for 3D artefact restoration [3], they mostly rely on matching 2D image projections against known reference images, an approach that cannot scale to blind geometric reassembly of unknown pottery. Consequently, applying population-based global optimization approaches directly to 3D geometric fracture constraints remains largely unexplored.

In this paper, we focus on reconstructing individual axially symmetric pots where all sherds belong to the same pot.¹ We build on SfS++ pipeline keeping its feature extraction functions as well as the pairwise matching stages, and replace its BAISER beam search with our proposed evolutionary algorithm system, SfS-GA. Our primary objective is to evaluate whether population-based evolutionary search can navigate through the combinatorial ambiguity presented by 3D artefact reassembly more effectively than greedy search methods. By benchmarking SfS-GA directly against SfS++, we compare reconstruction accuracy and computational cost to assess the viability of evolutionary algorithms in this domain.

The rest of this paper is organised as follows: Section II surveys related work concerning geometric matching, deep learning, parallel search, and genetic algorithms for fragment

¹Source code and evaluation pipeline are available at: <https://github.com/TahaZahid05/sfs-ga>

reassembly. Section III details our proposed GA methodology, including chromosome encoding, repair mechanisms, and fitness function components. Section IV describes the experimental setup and evaluation metrics. Section V presents our reconstruction results and discusses convergence dynamics along with limitations. Finally, Section VI concludes the paper.

II. LITERATURE REVIEW

A. Geometric Feature Matching Approaches

Initial computational pipelines attempt to solve 3D reassembly by relying heavily on geometric feature matching and extracting distinct curvature and roughness descriptors from fractured surfaces to guide pairwise alignment [4], [5]. While these classical methods proved highly effective for assembling generic solids with rich and irregular break topologies, they struggle significantly in the archaeological domain. Ceramic pottery typically shatters into fragments which are extremely thin and have sparsely featured fracture surfaces. When geometric methods rely solely on matching these ambiguous boundaries, they often generate many indistinguishable false-positive matches, causing sequential assemblies to fail due to error propagation. Hence, researchers in the domain noted that scalable and globally optimal assembling of fragmented symmetric objects is still a critical open challenge [8].

B. Data-Driven and Deep Learning Approaches

Recent advancements in the field have leveraged deep learning to automate feature extraction and global alignment. Learning-based models like PotNet [9] use 1D convolutional networks to predict the placement of a piece, while end-to-end architectures like Jigsaw [10], FRAISER [11], and PuzzleFusion++ [12] jointly learn segmentation, feature matching, and alignment using attention mechanisms, transformers, and diffusion models. While these approaches achieve impressive reconstruction accuracy on synthetic datasets like *Breaking Bad* [13], they show severe limitations when evaluated on real world archaeological pottery. Real ceramic sherds lack the distinctive bulky fracture topologies found in synthetic objects, rendering deep features unreliable and making data-driven models highly susceptible to false positive matches and poor generalisation.

C. Parallel Search Approaches

To address the challenges introduced by feature-sparse pottery, recent methods have utilised geometric priors, such as the pots’ axis of symmetry, to limit possible fragment placements while employing parallel search strategies to efficiently explore candidate alignments. Hong et al. introduced Structure-from-Sherds (SfS) [14], which relies on identifying a base fragment to guide the incremental reconstruction. SfS++ [2] improved upon this using its BAISER beam search to explore multiple assembly paths simultaneously. While SfS++ represents the current state of the art pipeline for real world axially symmetric pottery, the beam search strategy faces scaling challenges as the fragment count increases. For highly fragmented pots, distinguishing between true and false

configurations is extremely difficult in the early stages of assembly. To overcome these challenges, SfS++ must utilise a big search tree with very broad search branches creating a tradeoff between reconstruction accuracy and exponential computational runtime. Consequently, the struggle to maintain flawless performance across large pots remains.

D. Genetic Algorithms for Reconstruction

Genetic algorithms have proved to be highly successful in 2D reconstruction tasks, showing that they can navigate massive combinatorial spaces without committing early to greedy decisions. Sholomon et al. [15], [16] presents a GA-based solver for extremely large jigsaw puzzles, effectively handling unknown dimensions, pieces, and orientations using a specialized crossover procedure to merge correct sub-assemblies. Expanding to archaeological contexts, Sizikova and Funkhouser [17] applied GAs to global assembly of fragmented 2D wall paintings suffering from severe erosions, utilising fitness functions that reward dense match connections. These works highlight the scalability and strength of genetic algorithms in solving ambiguous 2D assembly problems.

To our knowledge, direct application of genetic algorithms on 3D pottery reassembly remains largely unexplored. While some efforts introduce evolutionary concepts to 3D artefact restoration [3], their underlying logic differs entirely from geometric reassembly. For example, Kashihara [3] proposed a GA that optimises 3D fragment positions by projecting them on 2D, mapping silhouettes and ORB image features with known “correct” reference images. Because this approach fundamentally requires prior knowledge of the target shape’s 2D projections to evaluate fitness, it cannot scale to the blind reassembly of unknown 3D pottery. Therefore, developing population-based evolutionary search which operates directly on the 3D geometric fracture constraints remains an open challenge that we tackle in this paper.

III. METHODOLOGY

A. Problem Formulation

The problem consists of two subproblems: (i) geometric feature extraction and matching of sherds, and (ii) the reassembly process [1].

Let $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$ denote the set of n pottery sherds, each represented as a 3D point cloud with associated surface normals. The goal is to find a set of rigid transforms $\{T_1, T_2, \dots, T_n\}$ and a connection graph G over $\mathcal{S}$ such that the assembled configuration reconstructs the true pot. Each edge (i, j) in G represents a physical contact between sherds s_i and s_j , with an associated relative transform T_{ij} .

Following SfS++ [2], we restrict our focus to axially symmetric pottery, where the rotational symmetry about the vertical axis constrains the search space.

B. Pipeline Overview

Our system utilises the geometric preprocessing stages of SfS++ and applies a genetic algorithm for the global assembly. The full pipeline is:

- 1) **Feature extraction:** Axis estimation, breakline extraction, and rim segmentation per sherd.
- 2) **Pairwise matching:** LCS-based matching produces a candidate pool $\mathcal{M}$ of pairwise matches.
- 3) **Pairwise pruning:** Geometrical consistency filtering reduces $|\mathcal{M}|$ to remove weak pairwise matches.
- 4) **Pair group construction:** Candidates in $\mathcal{M}$ are grouped by shard pair, forming $\mathcal{G} = \{G_1, G_2, \dots, G_m\}$ where each G_k contains all candidate matches between a specific pair of sherds.
- 5) **GA search:** A population-based search over chromosomes encoding which pair groups to accept and which candidate match to select in the accepted pair groups.

In our approach, the SfS++ system handles geometric feature extraction and pairwise candidate generation.

C. Chromosome Representation

A chromosome $\mathbf{c}$ is an integer vector of length $m = |\mathcal{G}|$, one gene per pair group. Gene $c_k \in \{0, 1, \dots, |G_k|\}$ encodes the match selection decision for pair group G_k :

$$c_k = \begin{cases} 0 & \text{reject all matches for this pair} \\ j & \text{accept the } j\text{-th candidate match in } G_k \end{cases} \quad (1)$$

D. Population Initialization

Each chromosome is initialised into a highly dense state. For every available pair group G_k that contains candidate matches, the algorithm actively assigns an initial candidate $j \geq 1$, effectively turning on every possible edge in the connection graph.

To select these initial candidates, they are assigned a density score:

$$\text{density}(j) = \frac{\text{inlier}(j)}{1 + \text{score}(j)} \quad (2)$$

where $\text{inlier}(j)$ represents the number of breakline correspondences falling within a distance threshold during registration, and $\text{score}(j)$ is the LCS dissimilarity metric.

We employ a weight-based roulette wheel selection procedure to sample the candidates. The probability of selecting candidate j in group G_k is strictly proportional to its density score, which is augmented by a dynamic pheromone level τ_k maintained over generations:

$$P(\text{select } j) = \frac{\text{density}(j) + \tau_k}{\sum_{j' \in G_k} (\text{density}(j') + \tau_k)} \quad (3)$$

Pair groups that successfully form structurally validated consensus loops during evaluation receive a pheromone boost, while unused or conflicting groups decay. This policy ensures the initial dense graph and subsequent mutations prioritise geometrically reliable, globally consistent candidates.

E. Chromosome Repair

The initialisation produces densely connected graphs filled with physical cycles. However, to reconstruct a single pot composed of N sherds, a valid assembly only requires $N - 1$ active edges forming a tree.

We tackle this by applying a chromosome repair function before fitness evaluation. The repair algorithm assesses the chromosome's active edges, sorts them by geometric density, and runs Kruskal's algorithm using a Disjoint Set Union (DSU). This process prunes the active edges down to exactly the best $N - 1$ pairs, forcing all remaining genes to 0. As a result, the repaired chromosome always represents a valid maximum spanning tree.

F. Fitness Function

The fitness function $F(\mathbf{c})$ evaluates the geometric quality of the assembly encoded by the repaired chromosome $\mathbf{c}$. The function aggregates four primary components to compute the total fitness:

$$F(\mathbf{c}) = F_{\text{inlier}} + F_{\text{consensus}} - F_{\text{overlap}} - F_{\text{comp}} \quad (4)$$

1) *Inlier Reward* (F_{inlier}): For each active match $c_k > 0$, the algorithm computes an inlier reward equal to the density metric of the selected candidate. This acts as the foundational geometric pull of the fitness function:

$$F_{\text{inlier}} = \sum_{k | c_k > 0} \text{density}(c_k) \quad (5)$$

2) *Consensus Reward* ($F_{\text{consensus}}$): The consensus reward assesses configurations where multiple sherds agree on a global transform, forming stable local clusters. For every pair of sherds (i, j) in the assembly that are *not* connected by an active gene, the algorithm searches their pair group G_{ij} for any currently inactive candidate transform $T_{\text{candidate}}$ that mathematically agrees with their assembled relative transformation $T_{\text{asm}} = T_j \cdot T_i^{-1}$.

To verify this agreement, we compute the relative difference $\Delta T = T_{\text{asm}} \cdot T_{\text{candidate}}^{-1}$ and check if both the translation error $e_t = \|\Delta \mathbf{t}\|_2 < \tau_t$ and the rotation error $e_R = \arccos\left(\frac{\text{Tr}(\Delta R) - 1}{2}\right) < \tau_R$ fall within strict thresholds. If a candidate satisfies these conditions, a flat scalar reward is accumulated:

$$F_{\text{consensus}} = \lambda_{\text{cons}} \sum_{(i,j) \notin \text{active}} \mathbb{I}(\text{consensus_match}(i, j)) \quad (6)$$

This rewards assemblies that are mutually supported by multiple unused candidate matches, effectively supporting the configuration through implicit loop closures.

3) *Overlap Penalty* (F_{overlap}): To penalise physically implausible interpenetration, we project every sherd into the assembled global space. We construct voxel-downsampled collision clouds (using a 3.0 mm grid resolution) for every sherd, excluding points within a narrow margin of the breakline edges to avoid penalising valid contacts. For any pair of non-adjacent sherds (i, j) in the assembly, we use kd-tree nearest-neighbour queries to compute a hit ratio:

$$h_{ij} = \frac{|\{p \in \mathcal{P}_i \mid \text{dist}(p, \mathcal{P}_j) < \epsilon_{\text{coll}}\}|}{|\mathcal{P}_i|} \quad (7)$$

We extract the maximum overlap ratio a sherd i experiences with any single other sherd j . If this maximum ratio exceeds

a minimum threshold ($h_{\min}$), a penalty is applied directly proportional to that maximum value:

$$F_{\text{overlap}} = \lambda_{\text{overlap}} \sum_{i | \max_j h_{ij} \geq h_{\min}} \left(\max_j h_{ij} \right) \quad (8)$$

4) *Component Penalty* (F_{comp}): Because the objective is to reconstruct a single vessel, the desired assembly is a single connected component. We apply a heavy penalty $\lambda_{\text{comp}} = 500.0$ for every disconnected sub-component C above the target of 1:

$$F_{\text{comp}} = \lambda_{\text{comp}} \cdot \max(0, |C| - 1) \quad (9)$$

This strongly drives the genetic algorithm toward a completely connected vessel.

G. Genetic Operators

1) *Selection*: We use tournament selection with a tournament size of 2 to provide low selection pressure and maintain sustained genetic diversity throughout generations.

2) *Crossover*: We employ a Breadth-First Search (BFS)-Partition Crossover operator. To generate a child chromosome, the operator performs a BFS traversal over the first parent's active connection graph, starting from a random shard. It inherits active match connections along the BFS tree until it fills exactly half of the target edges. The remaining edges required to form a complete spanning tree are then inherited directly from the second parent's active connections. This ensures the child obtains contiguous, topologically connected sub-assemblies from the parents.

3) *Mutation*: The algorithm utilises several mutation strategies executed probabilistically during reproduction. The base mutation rate p_m is set to 0.15 but switches to a hyper-mutation rate of 0.40 if the population's best fitness stagnates over successive generations, actively breaking the search out of a local optimum.

- **Neighborhood Mutate**: When triggered under the base mutation rate, this operator executes one of three local strategies with equal probability:

- 1) *Candidate Switch*: Selects an active pair group and swaps its current candidate match for another candidate within the same group, selecting the new candidate match via rank-based sampling on density.
- 2) *Pair Switch*: Deactivates a randomly selected active edge, which temporarily breaks the spanning tree into two disconnected components. It then identifies all inactive candidate edges that connect the two components and activates one to guarantee the assembly remains fully connected.
- 3) *Triangle-Builder*: Exploits the geometric structure of pottery by actively searching for triplet pairs that mutually agree with each other. It takes an active pair of sherds and searches for a third sherd where the available candidate matches form a geometrically consistent loop (i.e., their relative transforms compose correctly). When such an agreeing triplet

is found, the operator activates two of these three matching pairs which allows the chromosome to have geometrically consistent pairs without violating the spanning tree constraints.

- **Macro Mutate (Hyper-mutation)**: If the algorithm enters the $p_m = 0.40$ hyper-mutation state, a massive, sherd-centric structural mutation occurs. The algorithm selects a random target sherd and deactivates all its active connections. The algorithm then iterates through the fragmented components and reconnects them using the highest-density available candidate matches, which re-anchors the sherd in the reconstruction.
- **Mass Extinction**: To escape local minima, if the population's best fitness stagnates for 15 generations, a reset is triggered. The entire population is wiped out and replaced with completely new, randomly initialised spanning trees, except for the elite chromosomes which are preserved to anchor the new search phase. If best fitness stagnates for 120 generations, the GA terminates, and we take the best fitness as the final solution.

4) *Elitism*: The top $e = 10$ chromosomes are carried over directly to the next generation without modification, making sure the best solution found is never lost.

IV. EXPERIMENTAL SETUP

A. Dataset

We evaluate on the SfS++ benchmark dataset [2], which contains 3D scanned pottery sherds from ten different archaeological pots (labelled A through J, shown in Fig. 1). Each sherd is represented as a point cloud having surface normals, breakline points, and rim segmentation. We use the ground truth graph G^* and transforms $\{T_i^*\}$ to compute accuracy metrics.

B. Evaluation Metrics

We report the standard SfS++ accuracy metrics:

- **Sherd Accuracy (SA)**: The percentage of correctly placed sherds over the total number of sherds. A sherd is considered correctly placed if at least one of its edges is placed correctly within a rotation threshold and translation threshold of the ground truth transform.
- **Edge Accuracy (EA)**: The percentage of correctly identified pairwise connections over the total number of ground truth connections across the entire assembly.

C. Hyperparameters

Table I lists all GA hyperparameters used in our experiments.

D. Execution Environment

All experiments used the default parameters defined in the codebase, which uses deterministic random number generation (seeded with 42) to ensure exact reproducibility.

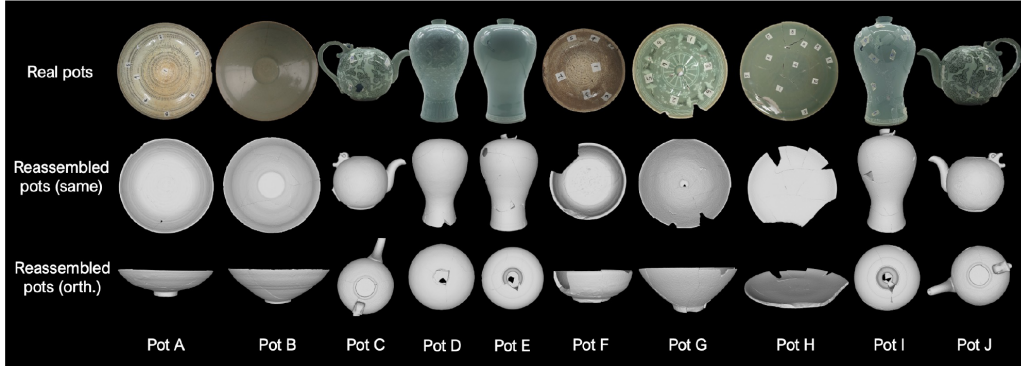


Fig. 1. The ten archaeological vessels (Pots A–J) from the SfS++ benchmark dataset, illustrating the real scanned pots alongside their ground truth assemblies (Image reproduced from Yoo et al. [2]).

TABLE I
GA HYPERPARAMETERS

Parameter	Value	Rationale
Population size	1000	Supports large search spaces
Early Termination	120 gen	Exits upon sustained stagnation
Elitism count	10	Preserves top 1% solutions
Base Mutation (p_m)	0.15	Steady local exploration
Hyper-Mutation (p_m)	0.40	Escapes local minima
Tournament size	2	Low selection pressure
λ_{comp}	500.0	Strong component penalty
λ_{overlap}	300.0	Base scale for overlap penalty
λ_{cons}	20.0	Consensus loop closure reward
τ_t	22.0 mm	Consensus translation tolerance
τ_R	0.22 rad	Consensus rotation tolerance
h_{min}	2%	Minimal overlap tolerance

V. RESULTS

A. Reconstruction Accuracy

Table II compares our GA against standard methods on the SfS++ benchmark. Our solution achieves 100% Sherd Accuracy and Edge Accuracy on Pots A, B, C, F, and G, matching up to the state of the art. On the remaining vessels (Pots D, E, H, I, and J), our GA produces competitive partial assemblies but lags behind SfS++. We analyse the reasons in Section VI.

VI. DISCUSSION

A. Fast Convergence on Manageable Datasets

For datasets characterised by a smaller number of fragments (e.g., Pots A, B, C, F, and G), the combinatorial search space remains manageable. On Pot A, for example, the algorithm rapidly explores this compact search space. It converges to the perfect ground-truth reconstruction almost instantly, as shown in the left panel of Figure 3. The best fitness score improves from an initial 554.6 at generation 0 to its global maximum of 592.6 by generation 3. This fast convergence shows that when lower sherd counts sufficiently bound the search space, the evolutionary search quickly finds the best configuration in few generations.

As shown in Figure 2, the final assemblies produced by our GA for Pots A and F match the archaeological ground truth,

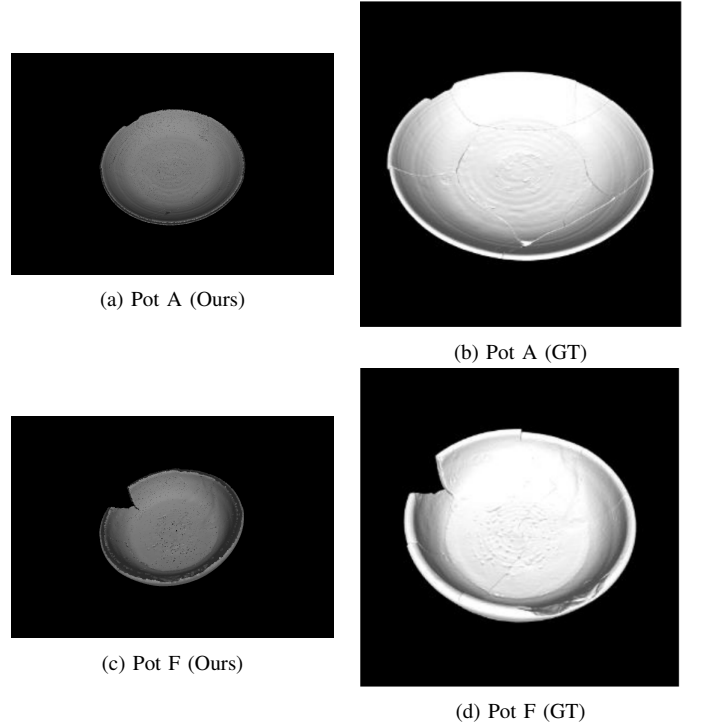


Fig. 2. GA reconstructions (left) vs. ground truth (right) for Pots A and F (100% SA and EA).

confirming that the 100% metric corresponds to a physically and visually consistent restoration.

B. Navigating Combinatorial Ambiguity in Complex Datasets

When datasets are highly ambiguous or massively fractured, the GA exhibits a distinctly different convergence profile (Figure 3). Using Pot E as a case study, the algorithm struggles to avoid local optima, characterised by prolonged plateaus in the best fitness graph. While the best fitness gradually improves from 717.6 at generation 0 to 1525.3 by generation 172, it stalls at this plateau for the remainder of the run. Notably, the average fitness exhibits severe periodic crashes corresponding to the GA’s mass extinction mechanism, which replaces a large

TABLE II
RECONSTRUCTION ACCURACY ACROSS MULTIPLE POTTERIES

Pottery (Sherds)	Jigsaw [10]		FRAISER [11]		PuzzleFusion++ [12]		SfS [14]		SfS++ [2]		SfS-GA (Ours)		
	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	SA	EA	RT(s)
Pot A (8)	25.0	6.7	0.0	0.0	62.5	46.7	0.0	0.0	100.0	100.0	100.0	100.0	452
Pot B (9)	22.2	6.7	22.2	6.7	33.3	20.0	100.0	100.0	100.0	100.0	100.0	100.0	367
Pot C (4)	0.0	0.0	100.0	40.0	50.0	20.0	100.0	100.0	100.0	100.0	100.0	100.0	574
Pot D (28)	—	—	—	—	—	—	82.1	74.2	82.1	83.3	75.0	67.4	33904
Pot E (31)	—	—	—	—	—	—	100.0	77.4	100.0	100.0	64.5	33.9	18942
Pot F (6)	0.0	0.0	33.3	11.1	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	133
Pot G (7)	42.9	20.0	0.0	0.0	100.0	100.0	0.0	0.0	100.0	100.0	100.0	100.0	124
Pot H (11)	0.0	0.0	0.0	0.0	0.0	0.0	81.8	76.5	90.9	88.2	45.5	29.4	1066
Pot I (27)	—	—	—	—	—	—	88.9	70.8	92.6	90.8	88.9	73.1	21961
Pot J (11)	0.0	0.0	0.0	0.0	0.0	0.0	18.2	4.8	72.7	76.2	63.6	23.8	3094

SA: Sherd Accuracy (%); EA: Edge Accuracy (%); RT: GA Runtime (sec).

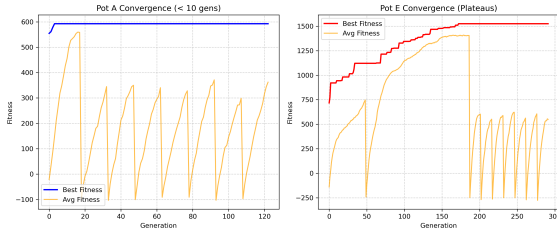


Fig. 3. Fitness convergence over generations. Left: Fast convergence on Pot A. Right: Prolonged plateauing on Pot E, with periodic crashes in average fitness due to mass extinction events.

portion of the population with newly randomised individuals every 15 generations of stagnation.

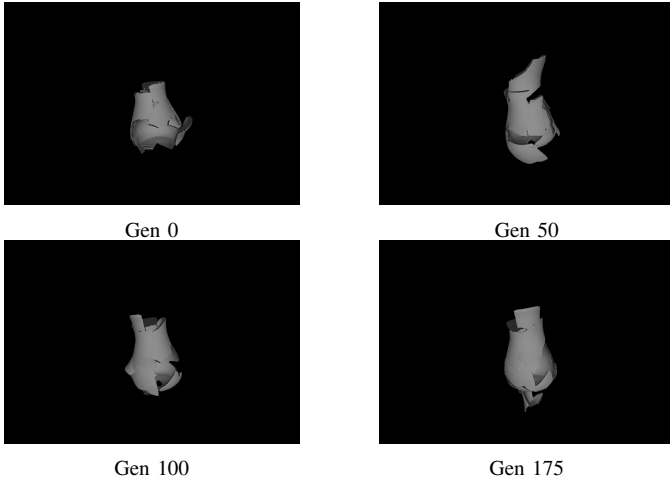


Fig. 4. Pot E assembly at key generations. The GA builds visually coherent sub-components but converges to a false-positive global shape.

During these plateaus, the GA constructs visually coherent but structurally flawed false-positive assemblies, as visualised in Figure 4. In fact, the final false-positive solution found by the GA scored a total fitness of 1525.31, which is higher than the ground-truth assembly score of 1405.15. This shows that the fitness landscape is deceptive as the GA converged to a configuration that was incorrect yet scored higher than the

ground truth configuration.

C. Upstream Limitations and Missing Ground Truth Pairs

Except for Pot E, where candidate pairs were present but the GA converged to a local optimum (Section VI-B), failure to achieve 100% accuracy on the remaining vessels (D, H, I, and J) stems from upstream matching. The initial pairwise matching and pruning failed to recover all ground-truth edges into the candidate pool. Without these pairs, complete reconstruction is mathematically impossible, forcing the GA to assemble the best-scoring configuration available from incomplete input. This highlights a key bottleneck that global assembly is strictly bounded by upstream pairwise recall.

D. Limitations and Future Work

The main limitation is the non-linear increase in runtime as fragment counts grow. While simpler vessels like Pot F and Pot G optimise in roughly two minutes (133s and 124s), Pot E (31 sherds) required 5.2 hours and Pot D (28 sherds) over 9 hours. Future work will focus on parallelising fitness evaluations to scale to larger collections. A second limitation is the dependence on upstream pairwise matching; future extensions should explore joint optimisation where the GA can actively query or refine candidate matches during assembly.

VII. CONCLUSION

We presented SfS-GA, a genetic algorithm system for the 3D reassembly of axially symmetric pottery. Our population-based search achieves 100% reconstruction accuracy on five benchmark vessels (Pots A, B, C, F, and G) and converges rapidly on compact search spaces. On complex vessels, combinatorial growth leads to runtime scaling challenges and deceptive local optima, with accuracy bounded by upstream pairwise recall. The results prove evolutionary optimisation as a viable framework for 3D archaeological reassembly while showing clear requirements for scalable full-vessel restoration.

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